API Automation Interview Question Bank

Deduplicated, ranked by importance and grouped by topic
Prepared for a Test Automation Engineer with 4+ years of experience

Questions in your original document	**299**
Exact duplicates removed (same question, different wording of punctuation or case)	**7**
Near-duplicates merged (same question asked in different words)	**171**
Unique questions in this PDF	**121**

Section 1: Critical (Must Prepare)	33 questions
Section 2: High Priority	57 questions
Section 3: Medium Priority	28 questions
Section 4: Low Priority / Quick Revision	3 questions

How to Use This Document

Questions are ordered by how likely they are to be asked and how much weight interviewers give them for a tester with 4+ years of experience. Work top to bottom: finish Section 1 before you spend time on Section 3 or 4. Inside each section the questions are grouped by topic, and within each topic the most important question comes first.

A red tag such as **[asked x3 in your bank]** means the same question showed up several times in your original document (in different wording). Repetition is a good signal that it is a frequently asked question.

Section	What it means
Critical (Must Prepare) 33 questions	Asked in almost every API automation interview at 4+ years. Be ready to explain the concept and to write the REST Assured code or Postman script on the spot.
High Priority 57 questions	Frequent practical and design questions. Answer with the approach, the code idea and a real example from your framework.
Medium Priority 28 questions	Occasionally asked, usually as follow-ups or when your resume mentions the topic.
Low Priority / Quick Revision 3 questions	Niche protocols and rare scenarios. Know the one-line idea.

What Interviewers Look For (4+ Years)

Topic	What the interviewer is testing
API & HTTP Fundamentals	Precise answers on HTTP methods, idempotency, status codes and REST vs SOAP.
REST Assured: Requests	Can you write GET/POST/PUT/DELETE requests, request/response specifications and parameters live?
REST Assured: Validation	JsonPath extraction, schema validation, nested JSON, response time, headers: beyond status code.
Authentication & Security	Basic vs OAuth2 vs token flows in code, JWT checks, and simple security tests (SQLi, XSS).
Framework Design	The core 4+ years topic: BaseTest, POJOs, config, utilities, data-driven, chaining, structure.
Postman & Newman	Scripting in the Tests tab, variables, chaining and running collections headless.
CI/CD & Reporting	How API tests run automatically and how failures are triaged and reported.
Scenario-Based	Approach questions: no documentation, pagination, async, versioning, dependencies.
Reliability & Performance	Flaky API tests, retries, timeouts, response-time checks.
Mocking & WireMock	Why and how you mock dependencies when the real service is unavailable.
Contract Testing	Consumer-driven contracts (Pact) vs schema validation.

Note: one entry ('API Testing in Java - Wrote code to validate request and response payloads using REST-assured') was a resume line; it is now part of the JSON validation question. Short fragments such as 'Request chaining' or 'Logging requests and responses' are phrased as full questions and merged with matching ones.

Table of Contents

Section 1: Critical (Must Prepare) (33 questions) . . . 5
API Testing & HTTP Fundamentals (5) . . . 5
REST Assured: Requests, Specifications & Parameters (7) . . . 5
REST Assured: Response Validation & JSON Handling (6) . . . 5
Authentication, Security & Rate Limiting (2) . . . 5
API Automation Framework Design (6) . . . 6
Postman Scripting & Newman (4) . . . 6
CI/CD & Reporting for API Tests (1) . . . 6
Scenario-Based API Testing (1) . . . 6
Reliability, Retry & Performance (1) . . . 6

Section 2: High Priority (57 questions) . . . 7
API Testing & HTTP Fundamentals (5) . . . 7
REST Assured: Requests, Specifications & Parameters (4) . . . 7
REST Assured: Response Validation & JSON Handling (6) . . . 7
Authentication, Security & Rate Limiting (3) . . . 7
API Automation Framework Design (9) . . . 7
Postman Scripting & Newman (9) . . . 8
CI/CD & Reporting for API Tests (3) . . . 8
Scenario-Based API Testing (12) . . . 8
Reliability, Retry & Performance (1) . . . 9
Mocking & WireMock (2) . . . 9
API Contract Testing (3) . . . 9

Section 3: Medium Priority (28 questions) . . . 10
API Testing & HTTP Fundamentals (2) . . . 10
REST Assured: Requests, Specifications & Parameters (3) . . . 10
REST Assured: Response Validation & JSON Handling (3) . . . 10
Authentication, Security & Rate Limiting (1) . . . 10
API Automation Framework Design (2) . . . 10
Postman Scripting & Newman (5) . . . 10
CI/CD & Reporting for API Tests (1) . . . 10
Scenario-Based API Testing (7) . . . 10
Reliability, Retry & Performance (1) . . . 11
API Contract Testing (3) . . . 11

Section 4: Low Priority / Quick Revision (3 questions) . . . 12
Authentication, Security & Rate Limiting (1) . . . 12
Scenario-Based API Testing (2) . . . 12

Topic Finder (all sections for one topic)

Use this if you want to study one topic at a time. Each number links to that topic's questions inside a section.

Topic	S1	S2	S3	S4	Total
API Testing & HTTP Fundamentals	5 Qs, p.5	5 Qs, p.7	2 Qs, p.10	-	**12**
REST Assured: Requests, Specifications & Parameters	7 Qs, p.5	4 Qs, p.7	3 Qs, p.10	-	**14**
REST Assured: Response Validation & JSON Handling	6 Qs, p.5	6 Qs, p.7	3 Qs, p.10	-	**15**
Authentication, Security & Rate Limiting	2 Qs, p.5	3 Qs, p.7	1 Qs, p.10	1 Qs, p.12	**7**
API Automation Framework Design	6 Qs, p.6	9 Qs, p.7	2 Qs, p.10	-	**17**
Postman Scripting & Newman	4 Qs, p.6	9 Qs, p.8	5 Qs, p.10	-	**18**
CI/CD & Reporting for API Tests	1 Qs, p.6	3 Qs, p.8	1 Qs, p.10	-	**5**
Scenario-Based API Testing	1 Qs, p.6	12 Qs, p.8	7 Qs, p.10	2 Qs, p.12	**22**
Reliability, Retry & Performance	1 Qs, p.6	1 Qs, p.9	1 Qs, p.11	-	**3**
Mocking & WireMock	-	2 Qs, p.9	-	-	**2**
API Contract Testing	-	3 Qs, p.9	3 Qs, p.11	-	**6**

S1 = Critical, S2 = High, S3 = Medium, S4 = Low priority.

Section 1: Critical (Must Prepare) (33 questions)

Asked in almost every API automation interview at 4+ years. Be ready to explain the concept and to write the REST Assured code or Postman script on the spot.

API Testing & HTTP Fundamentals (5)

1. What is API testing and why is it important?
2. Explain the HTTP methods used in RESTful APIs (GET, POST, PUT, PATCH, DELETE). **[asked x2 in your bank]**
3. What is idempotency in HTTP methods?
4. Explain HTTP status code categories and what 200, 201, 400, 401, 404 and 500 mean. **[asked x7 in your bank]**
5. What is the difference between SOAP and REST APIs? How do you test SOAP? **[asked x2 in your bank]**

REST Assured: Requests, Specifications & Parameters (7)

6. API automation: RestAssured vs Postman tests. When do you use which?
7. What is REST Assured? Explain its structure and components, and how you set it up in a project. **[asked x6 in your bank]**
8. Write a basic GET request using REST Assured. **[asked x4 in your bank]**
9. How do you send a POST request with a JSON body using REST Assured? **[asked x3 in your bank]**
10. What are Request Specification and Response Specification? How do you create and reuse them (base URI)? **[asked x9 in your bank]**
11. How do you handle query parameters and path parameters in REST Assured? **[asked x5 in your bank]**
12. How do you log request and response details (to console or a file)? **[asked x6 in your bank]**

REST Assured: Response Validation & JSON Handling (6)

13. How do you validate an API response? What are the most common validations? **[asked x2 in your bank]**
14. How do you validate the status code and response body with assertions (and multiple assertions in one request)? **[asked x7 in your bank]**
15. How do you validate JSON response data, extract values and use JsonPath (including checking a key is present)? **[asked x10 in your bank]**
16. How do you validate JSON schema in REST Assured? What is schema validation and why is it important? **[asked x7 in your bank]**
17. How do you validate response time? **[asked x4 in your bank]**
18. How do you test error handling, error responses, error codes and negative scenarios? **[asked x6 in your bank]**

Authentication, Security & Rate Limiting (2)

19. How do you handle authentication in REST Assured (basic, OAuth2, token-based, dynamic tokens)? **[asked x9 in your bank]**
20. What is the difference between authentication and authorization?

API Automation Framework Design (6)

21. How do you structure an API automation framework?
22. Implement a BaseTest class for API tests.
23. Write a POJO class for serialization/deserialization. How do you use Jackson/Gson and POJOs in API tests? **[asked x3 in your bank]**
24. How do you implement data-driven testing for APIs (TestNG DataProvider, JUnit, Excel/JSON)? **[asked x7 in your bank]**
25. How do you integrate REST Assured with TestNG or JUnit? **[asked x2 in your bank]**
26. How do you implement request chaining (extract a value and pass it to the next request)? **[asked x2 in your bank]**

Postman Scripting & Newman (4)

27. What are Collections, Environments and Variables in Postman? How do you use environment variables in scripts and import/export them? **[asked x5 in your bank]**
28. What is the difference between Pre-request Script and Tests in Postman?
29. How do you write a basic test script in Postman (status code, response time, JSON body assertions)? Which scripting language does it use? **[asked x4 in your bank]**
30. How do you chain requests in Postman (dynamic tokens, store and reuse values, capture header values, call one request from another, sequencing)? **[asked x7 in your bank]**

CI/CD & Reporting for API Tests (1)

31. How do you run API tests in CI/CD pipelines (Newman or REST Assured with Jenkins/Maven)? How do you trigger them after deployment? **[asked x6 in your bank]**

Scenario-Based API Testing (1)

32. How do you approach testing a new API when no documentation is available, or the requirements are incomplete and the behaviour changes often? **[asked x2 in your bank]**

Reliability, Retry & Performance (1)

33. How do you handle flaky or intermittent API tests? Implement retry logic for failed requests. **[asked x4 in your bank]**

Section 2: High Priority (57 questions)

Frequent practical and design questions. Answer with the approach, the code idea and a real example from your framework.

API Testing & HTTP Fundamentals (5)

34. What is the difference between API testing and UI testing frameworks?
35. What is the difference between URI, URL and endpoint?
36. What are headers in API requests and responses?
37. What is JSON and why is it preferred over XML?
38. What is the difference between smoke, regression and load testing in APIs?

REST Assured: Requests, Specifications & Parameters (4)

39. How do you perform PUT (with validation) and DELETE requests in REST Assured? Write sample code. **[asked x3 in your bank]**
40. How do you set headers and cookies, and extract cookies from a response? **[asked x4 in your bank]**
41. How do you test APIs with different content types (JSON, XML, form data)? **[asked x2 in your bank]**
42. How do you perform file upload and multipart form-data requests? **[asked x4 in your bank]**

REST Assured: Response Validation & JSON Handling (6)

43. How do you validate response headers, cookies and content type? **[asked x3 in your bank]**
44. How do you validate JSON arrays and lists containing specific values? **[asked x2 in your bank]**
45. How do you handle nested and deeply nested JSON responses? **[asked x4 in your bank]**
46. How do you handle dynamic data and dynamic payloads or JSON files? **[asked x2 in your bank]**
47. How do you compare two API response lists for equality ignoring order?
48. What are soft and hard assertions?

Authentication, Security & Rate Limiting (3)

49. How do you test API authentication (OAuth, JWT)? How do you validate JWT tokens? **[asked x2 in your bank]**
50. What is API security testing? How do you test broken authentication, SQL injection and XSS on an API? **[asked x6 in your bank]**
51. How do you test API rate limiting (throttling) and handle it in automation? **[asked x4 in your bank]**

API Automation Framework Design (9)

52. Design a config.properties management class and manage environment-based execution (dev, staging, prod). **[asked x4 in your bank]**
53. Implement the Singleton pattern for an API client.
54. Create a response validator utility and a reusable API helper class. **[asked x2 in your bank]**
55. Design endpoint management using Enums. Design a page object model for API endpoints. **[asked x2 in your bank]**
56. Create an end-to-end API test scenario.

57. How do you manage test data?
58. Explain TestNG groups for API test organization. How do you run tests in parallel? **[asked x2 in your bank]**
59. How do you integrate API and UI tests?
60. What are best practices for API automation? How do you keep API tests maintainable? **[asked x5 in your bank]**

Postman Scripting & Newman (9)

61. What is Newman? How do you run Postman tests headlessly and generate HTML reports? **[asked x4 in your bank]**
62. How do you send authentication credentials in Postman?
63. What is the Collection Runner?
64. How do you run data-driven tests and parameterize requests in Postman? **[asked x2 in your bank]**
65. How do you validate JSON schema and arrays in Postman? **[asked x3 in your bank]**
66. How do you run only selected tests, folders or requests from a collection? **[asked x2 in your bank]**
67. How do you handle loops, conditional logic, random data and console logging in Postman scripts? **[asked x4 in your bank]**
68. How do you retry failed requests in Postman scripts?
69. How do you pass query parameters in Postman?

CI/CD & Reporting for API Tests (3)

70. If an API test fails in CI/CD, how do you troubleshoot?
71. How do you pass dynamic environment variables in CI/CD?
72. How do you publish API test results to a dashboard?

Scenario-Based API Testing (12)

73. How do you test API pagination? **[asked x4 in your bank]**
74. How do you test sorting and filtering in APIs?
75. How do you test APIs that return asynchronous responses? **[asked x2 in your bank]**
76. How do you test APIs that depend on other services, and in a microservices architecture? **[asked x2 in your bank]**
77. How do you test APIs with third-party integrations?
78. How do you validate API caching behaviour? **[asked x2 in your bank]**
79. How do you test APIs under failure scenarios (including simulated network failures)? **[asked x2 in your bank]**
80. How do you test API concurrency issues?
81. How do you test APIs that behave differently for different users?
82. How do you handle API versioning in tests and ensure backward compatibility? **[asked x4 in your bank]**
83. How do you validate data integrity in API testing?
84. How do you test APIs with file uploads in Postman?

Reliability, Retry & Performance (1)

85. How do you handle timeouts, retries and error scenarios, and set a custom request timeout? **[asked x3 in your bank]**

Mocking & WireMock (2)

86. What is the purpose and what are the benefits of mocking in API testing? **[asked x2 in your bank]**
87. How do you mock an API with WireMock (set up and start the server, configure a response, call it with REST Assured, stop it)? **[asked x7 in your bank]**

API Contract Testing (3)

88. How do you validate API contracts? What is API contract testing and why is it important? **[asked x3 in your bank]**
89. What tools are used for contract testing? What is Pact and how does it work? **[asked x2 in your bank]**
90. What is the difference between schema validation and API contract testing?

Section 3: Medium Priority (28 questions)

Occasionally asked, usually as follow-ups or when your resume mentions the topic.

API Testing & HTTP Fundamentals (2)

91. What are the different types of APIs?
92. What is the difference between functional and non-functional API testing?

REST Assured: Requests, Specifications & Parameters (3)

93. How do you handle redirects in REST Assured?
94. How do you configure relaxed HTTPS validation?
95. How do you handle custom deserialization?

REST Assured: Response Validation & JSON Handling (3)

96. How do you parse XML responses and validate an XML schema? **[asked x3 in your bank]**
97. How do you handle optional fields in JSON validation?
98. How do you handle exceptions in REST Assured?

Authentication, Security & Rate Limiting (1)

99. How do you test data encryption in API/DB?

API Automation Framework Design (2)

100. How do you generate reports (Extent) for API tests? **[asked x2 in your bank]**
101. How do you version control API tests?

Postman Scripting & Newman (5)

102. What is a Monitor in Postman and how do you schedule collections? **[asked x2 in your bank]**
103. How do you mock APIs in Postman?
104. How do you manage large collections and keep environments in sync across teams? **[asked x2 in your bank]**
105. What are common mistakes in Postman testing?
106. How do you run Postman collections in parallel?

CI/CD & Reporting for API Tests (1)

107. How do you integrate Postman with version control (Git)?

Scenario-Based API Testing (7)

108. How do you test webhooks in Postman?
109. How do you test GraphQL APIs in Postman?
110. How do you test APIs with file downloads?
111. How do you test an API migration from monolith to microservices?
112. How do you test APIs with scheduled jobs?

113. How do you test localization in APIs?
114. How do you test APIs with complex JSON structures?

Reliability, Retry & Performance (1)

115. How do you perform load testing using REST Assured? How do you verify throughput and latency? **[asked x2 in your bank]**

API Contract Testing (3)

116. What are the key components of an API contract?
117. How do you ensure backward compatibility in contract testing?
118. What are the challenges of API contract testing?

Section 4: Low Priority / Quick Revision (3 questions)

Niche protocols and rare scenarios. Know the one-line idea.

Authentication, Security & Rate Limiting (1)

119. How do you test APIs that require digital signatures?

Scenario-Based API Testing (2)

120. How do you test gRPC APIs in Postman?
121. How do you test an API that returns streaming data?